

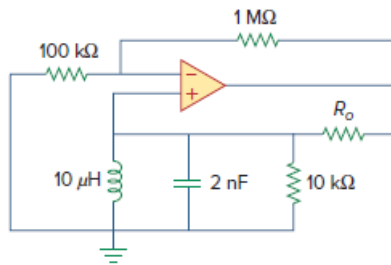
Problem

The oscillator circuit in Fig. 10.134 uses an ideal op amp.

(a) Calculate the minimum value of R_o that will cause oscillation to occur.

(b) Find the frequency of oscillation.

Figure 10.134:



Step-by-step solution

Step 1 of 9

Draw the following circuit diagram:

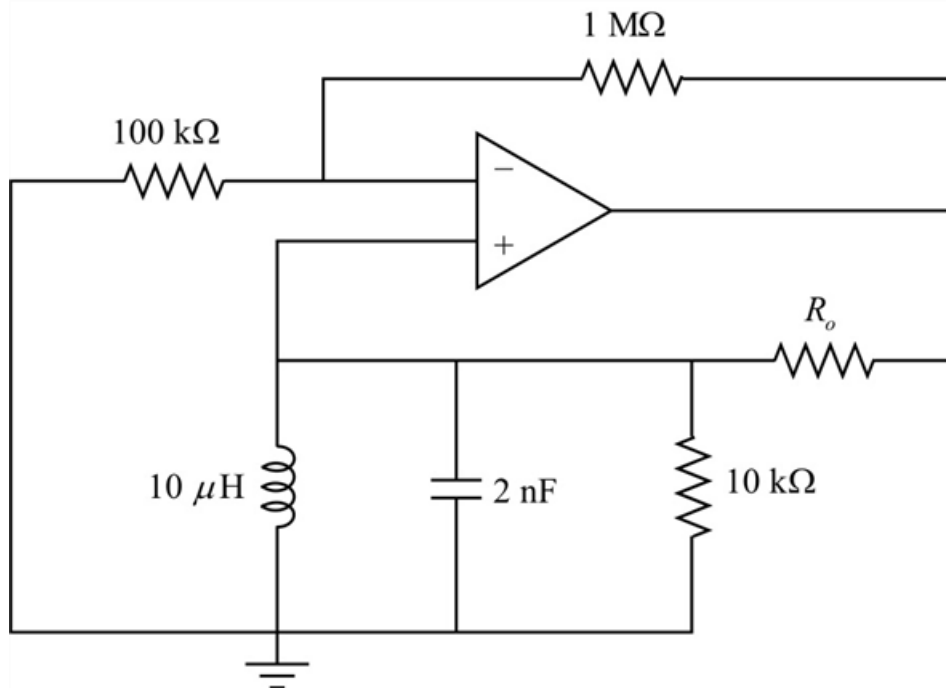


Figure 1

Step 2 of 9

Consider the following data:

The value of the resistor, R is $10\text{ k}\Omega$

The value of the inductor, L is $10\text{ }\mu\text{H}$

The value of the capacitor, C is 2 nF

The value of the feedback resistor, R_f is $1\text{ M}\Omega$

The value of the resistor, R_i is $100\text{ k}\Omega$

Step 3 of 9

(a)

Draw the modified circuit with notations:

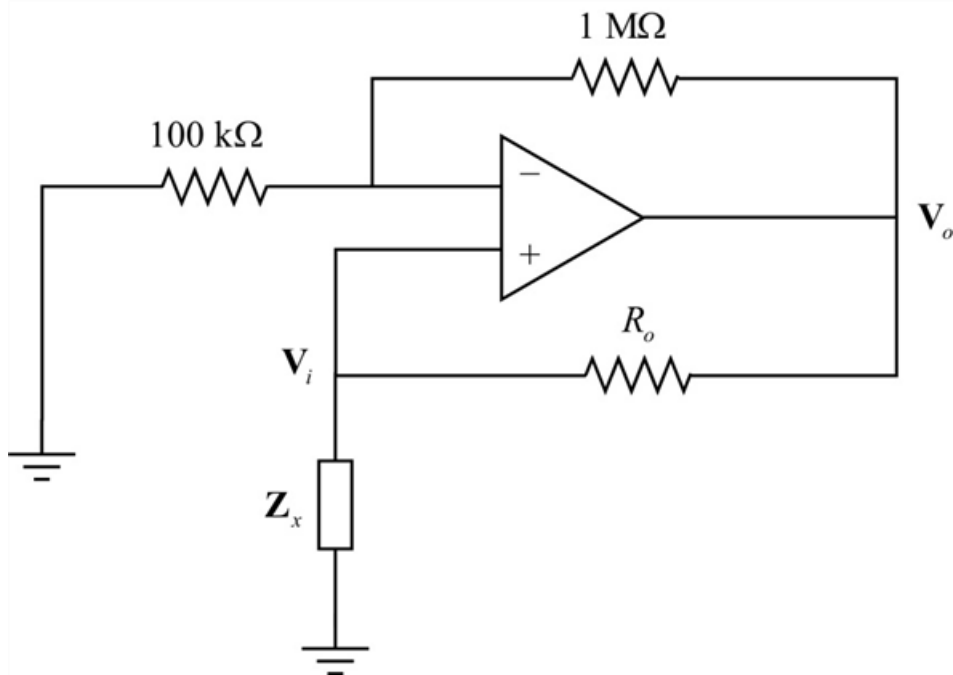


Figure 2

Step 4 of 9

Consider the following impedance Z_x :

$$\begin{aligned}\frac{1}{Z_x} &= \frac{1}{(j\omega L)} + (j\omega C) + \left(\frac{1}{R}\right) \\ Z_x &= \frac{1}{\frac{R - \omega^2 CLR + j\omega L}{j\omega LR}} \\ &= \frac{j\omega LR}{R(1 - \omega^2 CL) + j\omega L} \\ &= \frac{j\omega LR}{j[\omega L - jR(1 - \omega^2 CL)]} \\ &= \frac{\omega LR}{\omega L + jR(\omega^2 CL - 1)}\end{aligned}$$

Step 5 of 9

The condition for the circuit to be purely resistive is as follows:

$$\omega_o^2 CL - 1 = 0 \quad (\text{consider } \omega = \omega_o)$$

$$\omega_o^2 CL = 1$$

$$(2\pi f)^2 CL = 1$$

$$f = \frac{1}{2\pi\sqrt{LC}}$$

Step 6 of 9

Therefore, the impedance Z_x is as follows:

$$\begin{aligned} Z_x &= \frac{\omega LR}{\omega L + jR(1-1)} \\ &= \frac{\omega LR}{\omega L} \\ &= R \quad (\text{since } R = 10 \text{ k}\Omega) \\ &= 10 \text{ k}\Omega \end{aligned}$$

Step 7 of 9

Figure 2 is a non-inverting amplifier.

Calculate the gain of the amplifier.

$$\frac{V_o}{V_i} = 1 + \frac{R_f}{R_i}$$

Substitute $100 \text{ k}\Omega$ for R_i and $1 \text{ M}\Omega$ for R_f in $\frac{V_o}{V_i} = 1 + \frac{R_f}{R_i}$.

$$\begin{aligned} \frac{V_o}{V_i} &= 1 + \frac{10 \times 10^6}{100 \times 10^3} \\ &= 1 + 10 \\ &= 11 \end{aligned}$$

Apply voltage division rule for the circuit in Figure 2.

$$\begin{aligned} V_i &= \left[\frac{Z_x}{R_o + Z_x} \right] V_o \\ \frac{V_o}{V_i} &= \frac{R_o + Z_x}{Z_x} \\ R_o &= Z_x \left(\frac{V_o}{V_i} \right) - Z_x \\ R_o &= Z_x \left(\frac{V_o}{V_i} - 1 \right) \end{aligned}$$

Step 8 of 9

Substitute $10 \text{ k}\Omega$ for Z_x and 11 for $\frac{V_o}{V_i}$ in $R_o = Z_x \left(\frac{V_o}{V_i} - 1 \right)$.

$$\begin{aligned} R_o &= (10 \times 10^3)(11 - 1) \\ &= (10 \times 10^3)(10) \\ &= 100 \text{ k}\Omega \end{aligned}$$

Therefore, the value of the resistance R_o is 100 kΩ.

(b)

Calculate the frequency of oscillations f :

$$f = \frac{1}{2\pi\sqrt{LC}}$$

Substitute $10\ \mu\text{H}$ for L and $2\ \text{nF}$ for C in $f = \frac{1}{2\pi\sqrt{LC}}$.

$$\begin{aligned} f &= \frac{1}{2\pi\sqrt{(10 \times 10^{-6})(2 \times 10^{-9})}} \\ &= \frac{1}{2\pi\sqrt{200 \times 10^{-12}}} \\ &= 1.125 \times 10^6 \\ &= 1.125\ \text{MHz} \end{aligned}$$

Therefore, the frequency of oscillations is 1.125 MHz.